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Photoelectric effect: Back to basics

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Photoelectric effect: Back to basics^{a)}

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The measurement of Planck's constant by use of the photoelectric effect has long been a cornerstone experiment of the introductory modern physics laboratory. In this paper, the classic experiment is revisited in order to place more emphasis on the physics of the photoemission process. Toward this end, a measurement and theoretical analysis of the quantum yield of a commercial photocell are presented. Near the photoemission threshold, the quantum yield Y is shown to go as $(h\nu - \phi_c)^2$, where ϕ_c is the work function of the metal photocathode. Hence, ϕ_c can be determined from an experimental plot of $Y^{1/2}$ vs $h\nu$. In addition, justification is given for a procedure used for many years to determine the stopping potential from a plot of (photocurrent)^{1/2} versus applied bias voltage.

I. INTRODUCTION

One of the most popular introductory experiments in quantum physics is the measurement of Planck's constant (actually the ratio h/e) by use of the photoelectric effect.¹⁻⁴ The conventional approach is to focus monochromatic light onto the cathode of a commercial photocell and to measure the stopping potential V_{sp} as a function of frequency (usually, the five major lines of a Hg discharge lamp between 2-3 eV are chosen). That is, the photocell is reverse-biased so that the most energetic photoelectrons are just prevented from reaching the anode. The student verifies the Einstein photoelectric equation when the plot of V_{sp} versus frequency is found to be linear. The slope of the straight line then gives the value of h/e .

Not surprisingly, a great deal has been written about the photoelectric effect as a student experiment and the practical problems one encounters in its performance—principally, reverse photoemission from the anode which complicates the determination of stopping potentials.¹⁻¹³

The primary pedagogic point of the experiment is, of course, to demonstrate the quantization of the electromagnetic field. Unfortunately, most students who perform the classic photoelectric experiment are left unaware that the effect is of far more than historic interest. Over the last decade, considerable progress has been made in understanding and utilizing the photoelectric effect. As a result, photoemission spectroscopy has developed into a powerful technique to investigate the electronic band structure of solids.^{14,15} Electron spectroscopies based on the photoelectric effect have become important and growing subfields of modern solid-state physics.

In photoemission spectroscopy, monochromatic photons (usually with energy $h\nu$ in the uv or x-ray region) are incident on a solid surface. Electrons optically excited in the bulk which reach the surface with sufficient energy can escape into vacuum where their total number per absorbed photon, the quantum yield $Y(h\nu)$, as well as their distribution in energy $N(E, h\nu)$, can be measured. $Y(h\nu)$ and the latter energy distribution curves (EDC) are of interest because they are found to contain information about the detailed electronic band structure of the solid. Contrast this modern perspective with the classic student experiment in which (i) the variation of Y with photon energy is unimportant since the photocurrents obtained at different photon energies are normalized to the same saturation value and (ii) only electrons with one kinetic energy—the maximum

kinetic energy—are detected out of the total distribution $N(E, h\nu)$.

This article describes one way in which the classic photoelectric experiment can be amplified and made more instructive to advanced students, *viz.*, by measuring the quantum yield of the photocell. Although the EDC usually contain more information about the detailed electric band structure of a solid than does its quantum yield, EDC are much more difficult to measure experimentally and therefore not as suitable for the student laboratory. The experimental measurement of the photocell quantum yield will be described later in Sec. III. We present first a simplified theoretical analysis of the spectral dependence of the quantum yield, *i.e.*, the variation of the yield with photon energy. This analysis brings out aspects of the photoemission process not revealed in the conventional student experiment. The case of a metal photocathode will be treated since the photocell used in the present experiment (Leybold #55877) employed a potassium metal cathode.

II. THEORY OF PHOTOEMISSION

The generally accepted approach for interpreting photoemission data from solids is the use of the so-called "three-step model" developed by Spicer.¹⁶ The model assumes that the photoemission process can be broken down into the following three steps: (1) optical excitation in the bulk, (2) transport to the surface, and (3) escape into vacuum. In a metal, valence band electrons are first optically excited into higher-energy states in the conduction band by absorbed incident photons. Optical excitation will be closely related to the electronic structure of the solid. Secondly, the photoexcited electrons move through (diffuse randomly) the solid, and some fraction arrive at the surface. During the transport process, electrons can be scattered and lose energy—principally through electron-electron scattering. Thirdly, the electron must escape over the potential barrier at the surface if it is to be detected. As we will see, the probability of escape will depend on both the height of the surface potential barrier and on the angle of incidence between the electron and the surface.

Figure 1 indicates what is being measured in the photoemission experiment. The case of a free-electron-like metal having parabolic bands [density of states $D(E) \sim E^{1/2}$] is shown since this describes the valence band of potassium. All energies are measured with respect to the

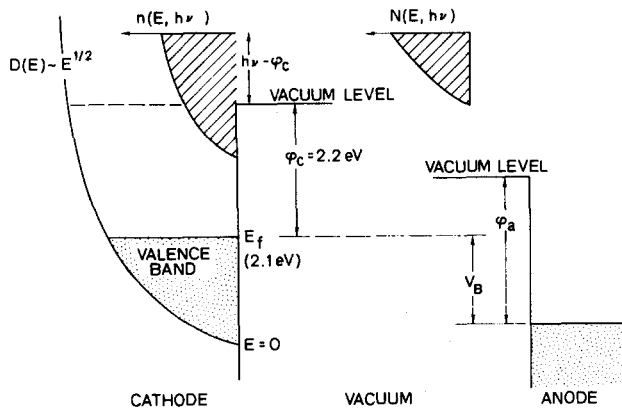


Fig. 1. Schematic of photoemission from a free-electron-like metal. Electron energies are measured with respect to the bottom of the cathode valence band. Values of ϕ_c and E_f are those for K taken from Ref. 17. $n(E, h\nu)$ represents the distribution of photoexcited electrons in the cathode and at the surface, while $N(E, h\nu)$ represents those electrons which have sufficient energy to overcome the surface barrier and escape into vacuum. The bias voltage V_B on the metal anode is shown large and positive, as is the case when the quantum yield is measured.

bottom of the valence band ($E = 0$). The values of the Fermi level ($E_f = 2.1$ eV) and the cathode work function ($\phi_c = 2.2$ eV) are those for potassium, taken from Ref. 17. The case in which the anode work function ϕ_a is greater than ϕ_c is shown. In Fig. 1, $n(E, h\nu)$ represents the energy distribution of photoelectrons in the metal and at the surface, while $N(E, h\nu)$ represents the distribution of those which are able to escape the surface barrier. As will be shown, $N(E, h\nu)$ is not simply given by $n(E, h\nu)$ truncated at energies below the vacuum level of the cathode.

The quantum yield is the total number of electrons contained in $N(E, h\nu)$. That is,

$$Y(h\nu) = \int_{E_0}^{E_m} N(E, h\nu) dE, \quad (1)$$

where $E_0 = E_f + \phi_c$ and $E_m = E_f + h\nu$. If $h\nu > E_0$ none of $N(E, h\nu)$ will lie below the vacuum level and the lower limit of integration is $E = h\nu$. In the present experiment $h\nu \lesssim 3$ eV and $E_0 \cong 4.3$ eV so that this was not the case. $N(E, h\nu)$ and $n(E, h\nu)$ are assumed normalized so that $Y(h\nu)$ has the units of photoemitted electrons/absorbed photon.

In order to collect all the electrons contained in $N(E, h\nu)$, i.e., to measure the quantum yield, the photocell is forward-biased (the anode biased positive with respect to the cathode) so that the vacuum level of the anode falls below the vacuum level of the cathode. For the case of $\phi_a > \phi_c$ shown in Fig. 1, the bias voltage which must be applied, V_B , is positive and greater than $\phi_a - \phi_c$. When $V_B = 0$, the Fermi levels of cathode and anode are aligned. When the stopping potential V_{sp} at a given photon energy is desired, as in the conventional experiment, the vacuum level of the anode is made to lie just above the most energetic photoemitted electron, i.e., V_B is negative and $V_B = \phi_a - h\nu \equiv V_{sp}$.

We now determine, subject to several simplifying assumptions, the spectral dependence of the quantum yield. The optical excitation rate from valence band states at an energy E to conduction band states at energy $E + h\nu$ (to conserve energy after one-photon absorption) is given by

Fermi's Golden Rule:

$$\text{excitation rate} \sim (2\pi/\hbar) |\langle i | \mathcal{H} | f \rangle|^2 \times D_{cb}(E + h\nu) D_{vb}(E), \quad (2)$$

where the free-electron-like conduction band (cb) and valence band (vb) densities of states $D(E) \sim E^{1/2}$ and the matrix element sandwiches the electron-radiation field Hamiltonian between initial and final states. As a first approximation, we ignore any momentum or energy dependence which may be contained in the matrix element¹⁸ and assume that the energy distribution of photoexcited electrons goes as $D_{vb}(E) D_{cb}(E + h\nu) \sim E^{1/2} (E + h\nu)^{1/2}$.

In this treatment of photoemission, we consider only those photoexcited electrons which are able to reach the surface without appreciable energy loss. A more detailed treatment of the photoemission process shows that the fraction of such electrons reaching the surface is given by the expression $f = (\alpha L)/(1 + \alpha L)$, where $\alpha(\text{cm}^{-1})$ is the absorption coefficient of the incident radiation ($1/\alpha$ gives the penetration depth of the radiation) and L is a characteristic distance between scattering events.¹⁴ In general, α is a function of photon energy, while L is a function of the electron's final-state energy in the cathode. Fortunately, for electrons photoexcited in potassium by photons in the energy range 2–3 eV, the value of this fraction is found to be fairly constant.¹⁹ This means that the distribution of electrons which reach the surface without appreciable energy loss will have the same shape as the original photoexcited distribution in the bulk. From Eq. (2), we have that $n(E, h\nu) \sim D_{cb}(E + h\nu) D_{vb}(E) \sim E^{1/2} (E + h\nu)^{1/2}$.

In order to determine the effect of the surface on $n(E, h\nu)$, consider an electron with momentum $\hbar\mathbf{k}$ and energy $E = (\hbar k)^2/2m$ traveling toward the vacuum surface at an angle θ with respect to the surface normal, where $\cos \theta = k_x/k$ (see Fig. 2). We assume that an electron can escape if the energy associated with its velocity component normal to the surface

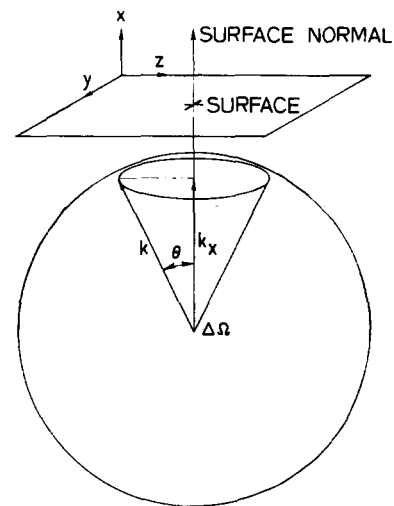


Fig. 2. An electron of momentum $\mathbf{k}$, making an angle of $\theta = \cos^{-1}(k_x/k)$ with respect to the surface normal, can overcome the surface barrier and escape into vacuum if its component of energy normal to the surface ($\hbar^2 k_x^2/2m$) is greater than the surface barrier E_0 . This condition leads to the escape of only those electrons whose momentum vectors lie within a cone of apex angle $\theta^* = \cos^{-1}(E_0/E)$ and solid angle $\Delta\Omega = 2\pi(1 - \cos \theta^*)$ where E is the electron energy.

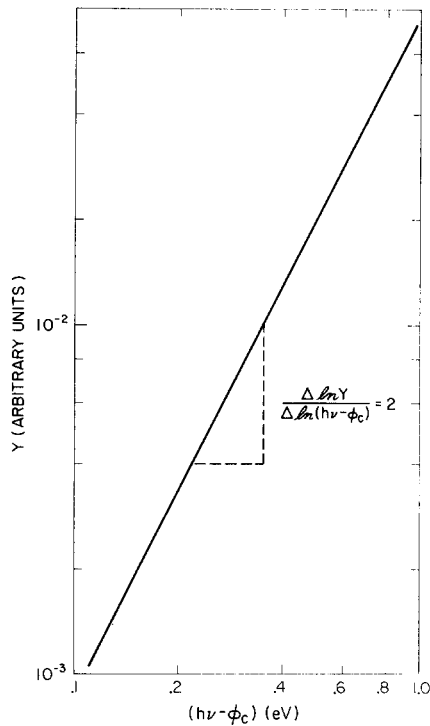


Fig. 3. Result of numerically integrating Eq. (6) is plotted as a function of $E_m - E_0 = h\nu - \varphi_c$ for $E_0 = 4.3$ eV. The slope of the log-log plot shows that $Y \sim (h\nu - \varphi_c)^2$.

exceeds the surface potential barrier. For a free-electron metal, the condition for escape is then given by

$$\hbar^2 k_x^2 / 2m = E \cos^2 \theta \geq E_f + \varphi_c \equiv E_0. \quad (3)$$

A critical angle θ^* therefore exists such that an electron with $\theta \leq \theta^*$ will escape but cannot for $\theta > \theta^*$. From Eq. (3), we see that $\cos \theta^* = (E_0/E)^{1/2}$. The probability $T(E)$ that an electron can escape is then the probability that it lies within the cone of solid angle $\Delta\Omega$ where

$$\Delta\Omega = \int_0^{2\pi} \int_0^{\theta^*} \sin \theta \, d\theta \, d\varphi = 2\pi(1 - \cos \theta^*). \quad (4)$$

If the electrons are excited such that their initial velocity distribution is isotropic in space, then $T(E)$ is simply the ratio of $\Delta\Omega$ to the total solid angle into which the electrons are initially excited. That is,

$$T(E) = \frac{\Delta\Omega}{4\pi} = \begin{cases} \frac{1}{2}(1 - \cos \theta^*) & E \geq E_0 \\ 0, & E < E_0. \end{cases} \quad (5)$$

When $E = E_0$, $T(E) = 0$. For electrons with energy much above the vacuum level, $T(E) \simeq 1/2$, since half of the electrons were initially headed away from the surface and, in the absence of scattering, will not return to escape.²⁰

Since $N(E, h\nu) = T(E)n(E, h\nu)$, the spectral dependence of the quantum yield can now be expressed as

$$Y(h\nu) = \int_{E_0}^{E_m} T(E)n(E, h\nu) \, dE \\ \sim \frac{1}{2} \int_{E_0}^{E_m} E^{1/2}(E + h\nu)^{1/2} [1 - (E_0/E)^{1/2}] \, dE. \quad (6)$$

We have evaluated the integral in Eq. (6) numerically

for $E_0 = 4.3$ eV and $h\nu = 2-3$ eV. The resulting log-log plot of Y vs $(h\nu - \varphi_c)$ in Fig. 3 shows that the spectral dependence of the yield is given by

$$Y^{1/2} \sim (h\nu - \varphi_c). \quad (7)$$

This square-root dependence arises primarily from the shape of the escape function $T(E)$ near threshold. (The threshold for photoemission occurs when $h\nu = \varphi_c$ corresponding to $E = E_f + \varphi_c = E_0$. This can be most easily seen by taking $n(E, h\nu)$ to be a constant. In this case,

$$Y(h\nu) \sim \int_{E_0}^{E_m} \left[1 - \left(\frac{E_0}{E} \right)^{1/2} \right] \, dE \\ = \int_0^{E_m - E_0} 1 - \left(\frac{E_0}{E_0 + \beta} \right)^{1/2} \, d\beta, \quad (8)$$

where $\beta = (E - E_0)/E_0 \ll 1$ near threshold. Thus, we can approximate Eq. (8) as

$$Y(h\nu) \simeq \int_0^{E_m - E_0} \frac{\beta \, d\beta}{2E_0} \sim (E_m - E_0)^2 = (h\nu - \varphi_c)^2. \quad (9)$$

If the escape function is assumed to be a step-function [i.e., $T(E) = 0$ for E below the vacuum level and $T(E) = 1$ for E above], the square-root relationship is *not* obtained. For a free-electron metal then, the spectral dependence of the quantum yield near threshold is determined by the shape of the escape function and not the density of states.²¹

III. MEASUREMENT OF THE QUANTUM YIELD

Experimentally, we wish to verify Eq. (7) that $Y^{1/2} \sim (h\nu - \varphi_c)$ and then determine the collector work function from the $h\nu$ intercept of such a plot. The reader is reminded that, in the conventional photoelectric experiment, the cathode work-function cannot be determined. Photon energies were chosen corresponding to the five major, visible Hg lines. In order to determine the quantum yield of the Leybold photocell, a standard detector of known quantum yield in this energy range is required. A large area (1.25 cm²) Schottky barrier Si photodiode (Model PIN-10) was purchased from UDT (United Detector Technology, Inc., Santa Monica, Calif.) for this purpose (cost $\approx$ \$60). The spectral response $\mathcal{R}$ of the Schottky barrier photodiode, as supplied by UDT, is shown in Fig. 4. The experimental setup employed was as follows: a Hg discharge lamp is focused with a quartz lens onto the entrance slits of a Bausch and Lomb 500-mm grating monochromator (model #33-86-45-48) which is supplied with a 600-line/mm plane reflection grating blazed at 3000 Å in first order. The image of the grating is focused by a quartz collective lens onto either the Leybold photocell or the PIN-10 photodiode. Both these tubes are sealed with glasses which do not transmit radiation for $h\nu \gtrsim 4$ eV, so that higher-order reflections of strong uv lines (e.g., the Hg line at 4.88 eV) will not contribute to the intensity of the first-order visible lines. Although a monochromator is convenient, it is not essential since only the sharp, visible lines of Hg are to be used. As in the traditional optical setup described elsewhere,⁸ these lines could equally well be dispersed by a prism and then individually aligned and focused onto the detector. The photocurrents I from the Leybold tube were measured with the tube strongly biased in the forward direction. On the anode, +60 V was found to be

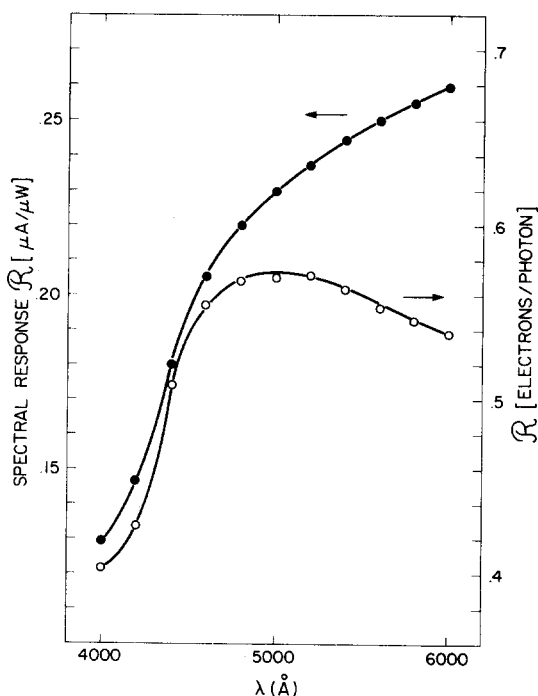


Fig. 4. Spectral response $\mathcal{R}$ of the UDT Schottky barrier photodiode model PIN-10 is shown from 4000 to 6000 Å. This response curve was supplied with the diode.

sufficient to saturate the collected photocurrents. These saturated photocurrents are, of course, much larger (1 to 2 orders of magnitude) and therefore easier to measure than the currents one has to deal with when the stopping potential is to be determined. The PIN-10 diode was used to measure the incident intensity of each Hg line. Photocurrents i were measured with a +5-V bias on PIN-10 so that the diode was operating in the desired photoconductive region. The photocurrents i from PIN-10 and I from the Leybold tube were each measured using a Keithly digital electrometer (model #616). Results obtained by a student (John Rizzo, class of 1979) are presented in Table I as well as the quantum yield Y of the photocathode in electrons/absorbed photon calculated as follows: $Y = I\mathcal{R}/[T(1 - R)i]$, where T is the transmission of the Pyrex glass sealing the cathode ($T \approx 100\%$ for $h\nu = 2\text{--}3\text{ eV}$) and R is the reflectivity of potassium ($R \approx 70\%$ and is nearly constant in this energy range; see Ref. 19). Hence, $Y \approx 3.3I\mathcal{R}/i$.

The spectral dependence of the quantum yield derived in Sec. II is borne out in the linearity of the $Y^{1/2}$ vs $h\nu$ plot

Table I. Photocurrents I and i from the Leybold tube and diode PIN-10, respectively. The response $\mathcal{R}$ of PIN-10 is taken from Fig. 4. The quantum yield of the Leybold potassium photocathode was calculated using $Y \approx 3.3I\mathcal{R}/i$.

$h\nu$ (eV)	λ (Å)	$\mathcal{R}$ (elec- trons/ photon)	$I \times 10^{-10}$ (A)	$i \times 10^{-6}$ (A)	$Y \times 10^{-3}$ (elec- trons/ absorbed photon)
2.15	5770	0.55	3.15	1.95	0.29
2.27	5461	0.56	7.9	2.55	0.57
2.52	4916	0.57	0.29	0.036	1.52
2.85	4358	0.49	40.5	1.8	3.64
3.06	4047	0.41	37.8	0.9	5.68

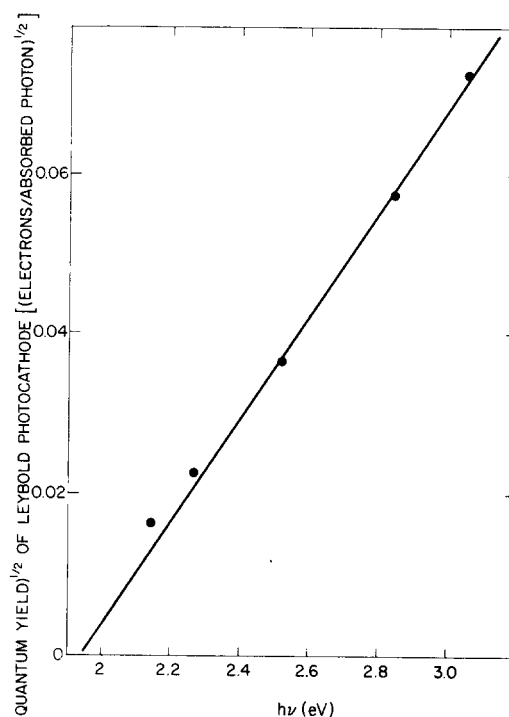


Fig. 5. Quantum yield Y of the Leybold tube's photocathode, calculated from the experimental student data in Table I, has been plotted as $Y^{1/2}$ vs $h\nu$ and verifies the spectral dependence of the quantum yield near threshold predicted by Eq. (7). The $h\nu$ intercept gives a value for the cathode work function of $\phi_c \approx 2.0\text{ eV}$.

of the student data shown in Fig. 5. The value of the cathode work-function obtained by the extrapolated intercept of the $h\nu$ axis is found to be $\phi_c = 1.95 \pm 0.05\text{ eV}$ —somewhat smaller than that of pure potassium (2.2 eV). The lower value may indicate contamination present at the cathode surface.

Under what conditions can we expect the measured $Y^{1/2}$ vs $h\nu$ plot to deviate from linearity? As mentioned in Ref. 21, if there is pronounced structure in the valence band density of states located close to the Fermi level, the behavior of a metal's quantum yield may deviate from the expected $Y^{1/2}$ dependence near threshold. Fortunately, this is not the case for the alkali-metal cathodes usually employed in the student experiment. Even for these cathodes, however, the square-root dependence is valid only for photon energies near the threshold of photoemission where the approximation $(E - E_0)/E_0 \ll 1$ used to evaluate Eq. (8) holds. For photocells using alkali-metal cathodes (e.g., Na, K), this approximation is found to hold for the photon energies used in the present experiment, *viz.*, $h\nu < 3\text{ eV}$.

The photoelectric experiment is sometimes performed using a photocell with a nonmetallic photocathode. For example, a Cs_3Sb cathode is used in the RCA phototubes 1P39 and 929. The quantum yield of such cells may exhibit a $Y^{1/2}$ dependence near threshold if the cathode density of states is relatively free of structure. In this case we expect the shape of the quantum yield near threshold to be determined primarily by the shape of the escape function $T(E)$, leading to the approximate linearity of $Y^{1/2}$ with photon energy. For the case of a nonmetallic cathode, however, the work function ϕ_c in Eq. (7) must be replaced by the photoemission threshold of the cathode—which is the sum of its electron affinity and band gap ($\sim 2\text{ eV}$ for Cs_3Sb).

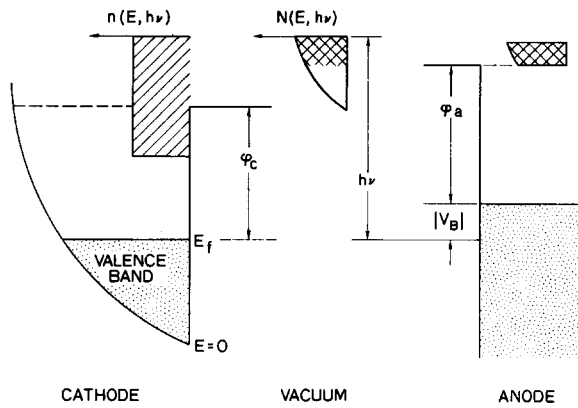


Fig. 6. Schematic of a photoemission experiment in which the bias voltage on the anode is such that only a fraction of the photoemitted distribution [the cross-hatched area of $N(E, h\nu)$ shown] will be collected. When $V_B = \phi_a - h\nu$, the vacuum level of the anode lies even with the highest energy photoelectron and V_B is then, by definition, the applied stopping potential V_{sp} . Analysis shows that, for bias voltages near the stopping potential, the anode photocurrent goes as $(h\nu - \phi_a + V_B)^2 = (V_B - V_{sp})^2$.

IV. DISCUSSION

Although, to our knowledge, not pointed out in the teaching literature, it has been common practice in photoemission research to determine the work function of metals by means of a plot of $Y^{1/2}$ vs $h\nu$.²² Such plots are based on a model for photoemission originally introduced by Fowler,²³ although the $Y^{1/2}$ dependence can be obtained by assuming only an isotropic distribution of excited electrons.²⁴

In this connection, several papers on the photoelectric experiment, which have appeared in this journal,^{5-7,13} are of interest. It has been noted in these papers that the square root of the photocurrent is found to be nearly linear with increasing negative bias potential within about 1 V of the stopping potential. Although no explanation for this relationship (apparently first reported by Lukirsky and Prilezaev²⁵) has been offered, its use has been recommended in order to determine V_{sp} . The procedure has been to plot $(\text{photocurrent})^{1/2}$ vs V_B . V_{sp} is then taken, arbitrarily, to be the point where the straight line obtained intercepts the V_B axis.

We now show that this useful procedure follows from an analysis similar to that already given in Sec. II for the quantum yield. Figure 6 shows a distribution of photoexcited electrons $n(E, h\nu)$ taken, for simplicity, to be constant. In this case, $N(E, h\nu) = n(E, h\nu) T(E) \sim T(E)$. As the vacuum level of the anode is raised relative to that of the cathode (i.e., the bias voltage V_B is made more negative), a smaller fraction of $N(E, h\nu)$ will be collected. The photocurrent collected is represented by the cross-hatched area of $N(E, h\nu)$ shown in Fig. 6. Unlike the case of *escape* from the cathode treated earlier, *collection* by the anode involves no critical angle of incidence. An electron is collected if its energy lies above the vacuum level of the anode and is not collected if its energy lies below. The collected photocurrent for a given bias voltage $V_B < 0$ is then given by

$$\begin{aligned} \text{photocurrent} &\sim \int_{E_b}^{E_m} N(E, h\nu) dE \\ &\sim \int_{E_b}^{E_m} T(E) dE \sim (E_m - E_b)^2, \end{aligned} \quad (10)$$

where the integral of $T(E)$ has been evaluated as before [see Eqs. (8) and (9)]. From Fig. 6, it is seen that the integral in Eq. (10) must be cut off at the vacuum level of the anode which lies $E_b = (E_f + \phi_a - V_B)$ in energy above the valence band minimum of the cathode ($E = 0$). Hence, the photocurrent $\sim (E_m - E_b)^2 = (E_f + h\nu - E_f - \phi_a + V_B)^2 = (h\nu - \phi_a + V_B)^2$. This result demonstrates that a plot of $(\text{photocurrent})^{1/2}$ vs V_B will be linear and intercept the V_B axis at $V_B = \phi_a - h\nu$, i.e., the stopping potential V_{sp} . As mentioned, this useful result has been applied but not justified in previous treatments of the photoelectric experiment. Hence, a plot of $I^{1/2}$ vs V_B enables the stopping potential to be determined. Knowing the stopping potential at a given $h\nu$ enables the anode work function to be determined, $\phi_a = h\nu + V_{sp}$. As we have shown earlier, a plot of $Y^{1/2}$ vs $h\nu$ enables the cathode work functional ϕ_c to be determined as well.

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¹⁸If the ground and excited electronic states in the metal are described by Bloch functions, optical transitions are allowed only if the initial and final states have the same wave vector, i.e., the matrix element vanishes for $\mathbf{k}_i \neq \mathbf{k}_f$. However, it has been found that the experimental optical data for a large number of materials can be understood by ignoring any $\mathbf{k}$ dependence in the matrix element (i.e., by assuming that $\mathbf{k}_i = \mathbf{k}_f$ is not an important selection rule) and demanding only that the energy is conserved so that $E_f = E_i + h\nu$. The reader is referred to Ref. 14 for a more complete discussion of this topic.

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²⁰The occurrence of electron-phonon scattering in a real solid can result in values for $T(E)$ above $1/2$. Phonon scattering is nearly elastic in metals at room temperature (energy loss per collision 10^{-2} - 10^{-3} eV) so that

an electron initially headed away from the surface could, by momentum transfer via phonon collisions, scatter back toward the surface with little energy loss.

¹In some metals, pronounced structure in the valence band density of states may exist (e.g., in Cu a high density of d states exists approximately 2 eV below E_F). If such structure is located close enough to the top of the valence band (Fermi level), the expected square-root spectral

dependence of Y near threshold may not hold. This is not a problem with the alkali-metal cathodes usually employed in the photoelectric experiment.

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